MILITARY SPECIFICATION

CRANES, TRUCK-MOUNTED, CONTAINER HANDLING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial diesel-engine-driven, truck-mounted, pneumatic-tired, full revolving cranes capable of lifting and placing loads mainly for container handling operations.

1.2 Classification. Cranes shall be one of the following types as specified (see 6.2):

Type I - 140 Ton
Type II - 250/300 Ton

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.

FSC 3950

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-G-10924	- Grease, Automotive and Artillery.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

STANDARDS

MILITARY

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

POWER CRANE AND SHOVEL ASSOCIATION

Mobile Power Crane and Excavator Standards (PCSA Standard No. 1)

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 E. Wisconsin Ave., Milwaukee, WI 53202.)

US DEPARTMENT OF LABOR

Occupational Safety and Health Act (OSHA) Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15906.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION

Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B30.5 - Standard Safety Code for Crawler, Locomotive and Truck Cranes.
Z53.1 - Marking Physical Hazards and Identification of Equipment.

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNDERWRITERS LABORATORIES

UL-62 - Flexible Cord and Fixture Wire.

(Application for copies should be addressed to the Underwriters Laboratories, 333 Pfingsten Road, Northbrook, ILL 60062.)

3. REQUIREMENTS

3.1 General. The crane shall consist essentially of the revolving superstructure, complete with front-end operating equipment mounted on a pneumatic rubber tired, heavy duty, truck type carrier. As required by law for commercial sales, the crane shall meet all applicable requirements of the Occupational Safety and Health Administration (OSHA) Standards, the Environmental Protection Agency (EPA) Emission Standards and Department of Transportation (DOT) Federal Motor Vehicle Safety Standards and Federal

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Motor Carrier Safety Regulations. Also, the crane shall meet all requirements set forth for compliance with the Power Crane and Shovel Association Standard Number 1. For clarification and illustration, certain requirements of these regulations have been repeated herein. This does not release the contractor from meeting the requirements that are not specifically stated herein. The crane and all of its components shall be the current commercial production model of a standard product and, except for product improvements, shall have demonstrated industry acceptability by having been manufactured, marketed and sold to commercial users for the same or similar application for at least 1 year prior to the opening of the bids.

3.2 First article. Unless otherwise specified (see 6.2), the contractor shall furnish one complete crane for first article inspection and test. The first article shall be identical to the other production cranes. When specified (see 6.2), the first article shall remain at the contractor's plant as a production sample, and shall be the last crane delivered on the contract. The first article shall be restored to a like new condition which meets all requirements of this specification, including tuneup, paint or repainting, prior to delivery. Any of the first article tests which the contractor has previously performed to meet the requirements of the American National Standards Institute (ANSI) Safety Code B30.5, PCSA Standard No. 1, or for a previous military contract, need not be performed providing these tests either equal or exceed the test specified in Section 4 of this specification, and have been adequately documented. A certified copy of the test report of the previously performed test shall be included as part of the first article test report.

3.3 Material. All material used in the manufacture of the crane shall be of the same quality as that being used by the contractor in his production cranes for commercial market sales. Material shall be free from any defects or imperfections that would affect the crane's operation or serviceability (see 6.6).

3.4 Performance. The crane shall be capable of performing both independently and simultaneously, the operational functions of hoisting, swinging, and booming. The crane shall be capable of traveling under its own power when fully equipped for lifting crane operation including a boom of not less than 130 feet over the type of rough terrain normally encountered at commercial construction sites.

3.4.1 Lifting capacities. The crane's standard commercial rated lifting capacities shall be not less than those shown in Table I. The crane shall be rated in accordance with ANSI B30.5. The contractor's standard commercial load chart shall be provided.

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TABLE I

LIFTING CAPACITIES
(Using Outriggers Fully Extended)

TYPE I

Boom Length (ft)	Radius (ft)	Capacity (lb), over side & rear
90	40	70,000
	45	63,000
	50	54,750
	60	42,750
	70	34,750
	80	29,000

TYPE II

Boom Length (ft)	Radius (ft)	Capacity (lb) over the side	Capacity (lb) over the rear
130	50	124,000	158,000
	90	52,000	70,000
	100	45,000	60,000

3.4.2 Operational environment. The crane shall be capable of starting and maintaining constant performance characteristics in any stabilized temperature from +120°F to -20°F.

3.4.3 Operational power requirements. The crane shall be capable of simultaneously lifting and rotating the rated load that produces the maximum load moment and the maximum single line pull at the rated radius for that load as designated in the contractor's load chart. The crane, with necessary boom sections and accessories removed to meet highway weight limitations, shall be capable of attaining a maximum highway travel speed on the level of not less than 30 miles per hour. The crane, complete with basic crane boom and required accessories, shall be capable of attaining a swing speed of not less than 3 revolutions per minute. The crane, completely equipped with basic boom and required accessories, shall be capable of negotiating a 15 percent grade on a surface with a rolling resistance of 40 lbs/ton. The crane equipped with a 130 foot boom shall be capable of travel at reduced speed over a hard surface roadway.

3.5 Construction. The crane shall be so designed and constructed that it can be conveniently disassembled into major components which come within the size and weight limitations for rail, highway and marine transportation. No component shall weight more than 65 tons. All parts, components, and attachments shall be identical for all cranes furnished. No deviation will be accepted without prior written approval of the contracting officer.

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3.5.1 Crane structure stress requirements. All of the structural members, the boom and the supporting components of the crane shall meet or exceed the strength margins specified in SAE J987.

3.5.2 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service which will be accomplished with the use of conventional, general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the crane.

3.5.3 Safety. The crane shall conform to ANSI B30.5. In addition, the crane shall comply with all Federal regulations which govern equipment and personnel safety requirements in effect on the date of manufacture.

3.5.3.1 Strike-hazard. Crane cab and portions of the crane that extend out and over the main chassis when being slewed shall have the lower area and backend color coded in accordance with ANSI Z53.1 to denote a strike hazard to personnel on the ground. Colors shall be reflective for night operations.

3.5.3.2 Backup-alarm. A backup alarm shall be provided on the crane and shall comply with SAE J994.

3.5.3.3 Noise level. If the noise level of the crane (excluding horns and warning alarms) exceeds 85 dB(A) when measured in accordance with 4.3.10, warning plates shall be permanently affixed to the crane in a conspicuous, protected location and shall be clearly visible and legible to all personnel exposed to the excessive noise levels. The warning plates shall read: CAUTION -- HEARING PROTECTION REQUIRED, or CAUTION -- HEARING PROTECTION REQUIRED WITHIN X FEET. The plate shall have a yellow background with black lettering and shall be of corrosion resistant material. The noise level of the crane shall not exceed 90dB(A) when measured in accordance with 4.3.11.

3.5.4 Human engineering. The design of the crane shall allow for operation by the 5th to 95th percentile male and female operator, as defined in SAE J833, suitably dressed for the environment. The location of controls shall comply with SAE J898.

3.5.5 Electromagnetic interference. The electromagnetic interference characteristics of the crane shall meet the recommended limits specified in SAE J551 when measured in accordance with SAE J551.

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3.6 Carrier. The carrier shall be of the multiaxle truck type and shall provide the required mobility and stability to the superstructure in all aspects of crane operations.

3.6.1 Outriggers.

3.6.1.1 Standard outriggers. Hydraulic outriggers shall be furnished, and shall be provided with means for dependent vertical and horizontal movement of the outrigger floats. The contractor's standard outrigger floats shall be furnished and shall be securely stored on the crane when not in use. The outrigger boxes shall have provisions for easy removal from the carrier chassis.

3.6.1.2 Front and rear auxiliary outriggers. In addition to standard outriggers, the Type I crane shall be provided with one front auxiliary and the Type II crane shall be provided with one front and two rear auxiliary outriggers complete with floats to provide for stability for operations over the front and rear of the carrier.

3.6.1.3 Low ground bearing pressure outrigger floats. In addition to the standard and auxiliary floats, the crane shall be provided with an outrigger float system for crane operation on surfaces with low load carrying capabilities. The system shall consist of a multiple float system with structural members for installation on the crane outrigger beams or special outrigger beams. Unless otherwise specified (see 6.2), the Type I crane shall be provided with a system consisting of not less than three floats for each outrigger beam end providing a total ground bearing area of not less than 8100 square inches. When specified for the Type I crane (see 6.2) and unless other specified for the Type II crane (see 6.2) the ground pressure of the complete crane with load shall not exceed 12 pounds per square inch (psi) when operating with rated loads. The contractor shall furnish his engineering calculations indicating compliance with this requirement.

3.6.2 Tires, rims and wheels. All wheels, rims and tires shall be the same size and type. Tires, rims and wheels shall conform to the recommendations of the Tire and Rim Association applicable to the type of application and load. A spare rim and tire shall be furnished for each crane.

3.6.3 Carrier brake system. Service and parking brakes conforming to all applicable requirements of Federal Motor Vehicle Safety Standards, Federal Motor Carrier Safety Regulations and ANSI B30.5 shall be provided. In addition, the parking brake shall be capable of holding the crane equipped with the basic boom on a grade of at least 15 percent when facing in either direction.

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3.6.4 Steering. The crane shall be equipped with hydraulic power-assist steering.

3.6.5 Carrier lighting and associated electrical system. A 12 or 24 volt lighting and electrical system conforming to Federal Motor Vehicle Safety Standard No. 108 shall be provided. Electrical components shall be suitable for use in a marine atmosphere.

3.6.6 Cab. The contractor's standard commercial carrier cab shall be furnished, and shall conform to Federal Motor Vehicle Safety Standards and Federal Motor Carrier Safety Regulations with respect to windshield and window glazing, windshield wiper, heater and defroster, operator's controls, seats and seat belts and cab doors. Provisions for adequate ventilation shall be provided and shall be easily adjustable by the operator. Hand holds and steps shall be provided to facilitate access to the cab in accordance with ANSI B30.5.

3.6.7 Carrier accessories. All carrier accessories required for safe operation in conformance to the DOT Motor Carrier Safety Regulation shall be provided. A fire extinguisher is not required.

3.7 Diesel engines. For the purposes of selecting the engines the crane shall be classified as a Class I end item in accordance with MIL-STD-1410. The diesel engines including all accessories and components shall be Class I of MIL-STD-1410 unless otherwise specified herein. The engines shall start in any temperature above minus 20°F. Starting aids shall be provided and may be either electric glow plug or ether pressure primer. When an ether priming system is furnished, it shall be of the measured shot type with a storage capacity of not less than 20 fluid ounces. The engines shall conform to the requirement of the Environmental Protection Agency (EPA) Federal Emission Standards.

3.7.1 Carrier engine

3.7.1.1 Carrier power and speed rating. The power required to permit the crane to attain and maintain a highway speed of not less than 30 mph and to permit the completely assembled crane to negotiate a 15 percent grade shall not exceed the engine manufacturer's intermittent horsepower rating.

3.7.1.2 Instruments and instrument panel. An instrument panel shall be furnished, the instruments shall be illuminated and shall include at least an engine oil pressure gage, an engine coolant temperature gage, an ammeter or voltmeter, a fuel level gage, and a speedometer that incorporates an odometer. An hourmeter shall be provided, shall be either instrument panel or engine mounted and need not be illuminated.

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3.7.1.3 Carrier fuel tank. A fuel tank conforming to Federal Motor Carrier Safety Regulations and of sufficient capacity for a driving range of not less than 200 miles shall be provided. If the tank is located in such a position that it may be walked on, it shall be covered with an antiskid type material.

3.7.1.4 Carrier starting system. An electric cranking system shall be furnished. The carrier shall be equipped with starting facilities to allow the operator to start the engine while seated in the normal operating position in the carrier cab.

3.7.1.5 Carrier storage batteries. Storage batteries furnished shall be of the lead-acid type conforming to SAE J537 or J930 and shall be housed in accordance with the Federal Motor Carrier Safety Regulations. Batteries shall be mounted so as to provide easy access for servicing.

3.7.1.6 Carrier battery charging. Provision shall be furnished to maintain the battery in the fully charged condition.

3.7.2 Revolving superstructure engine.

3.7.2.1 Superstructure power and speed rating. The superstructure engine, at full-load, governed speed, shall develop intermittent power equal to or exceeding the power required for all craning operations.

3.7.2.2 Instruments and instrument panel. An instrument panel shall be furnished, the instruments shall be illuminated, and shall include at least an engine oil pressure gage, an engine coolant temperature gage, an ammeter or voltmeter, and a fuel level gage. An hourmeter shall be provided, shall be either instrument panel or engine mounted and need not be illuminated.

3.7.2.3 Superstructure fuel tank. A fuel tank conforming to ANSI B30.5 and of sufficient capacity for not less than 6 hours of operation shall be furnished.

3.7.2.4 Superstructure starting system. An electric cranking system shall be furnished. The crane shall be equipped with starting facilities to allow the operator to start the engine while seated in the superstructure cab in the normal operating position.

3.7.2.5 Superstructure storage batteries. Storage batteries furnished shall be of the lead-acid type conforming to SAE J537 or J930 and housed in suitable battery carriers. Batteries shall be mounted so as to provide easy access for servicing.

3.7.2.6 Superstructure battery charging. Provision shall be furnished to maintain batteries in the fully charged condition.

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3.8 Revolving superstructure. The superstructure shall consist essentially of a cab enclosing the machinery and operator's position, mounted on a revolving platform. When drums are not enclosed in the cab, provision shall be made for weather protection of the clutches, brakes and gears. The crane shall be capable of self-installation of counterweights.

3.8.1 Cab. The contractor's standard commercial cab shall be furnished and shall conform to ANSI B30.5. For Type II cranes, the operator's cab shall be of the elevated type and shall provide the crane operator a line of vision originating at not less than 30 feet above the crane working platform when in the seated position. A power driven windshield wiper shall be furnished with a control switch within easy reach of the operator. A commercial type heater and defroster shall be furnished which will direct a steady flow of heated air to the operator's feet and capable of defrosting at least 75 percent of the front window area.

3.8.2 Gantry. A gantry, A-frame, or mast, of sufficient strength to withstand stresses induced by lifting the maximum length boom from the ground to operating position, shall be furnished. The gantry, A-frame, or mast shall be foldable, retractable or lowerable without requiring the use of an outside power source, in order to reduce traveling or transport cube.

3.8.3 Turntable.

3.8.3.1 Supporting rollers. The revolving superstructure shall be supported by large fixed rollers, rollers in pairs on tandem bogies, a live ball, or by a roller ring. The rollers shall be heat-treated steel and shall be accurately machined. Fixed rollers or rollers in pairs on tandem bogies shall be fitted with antifriction bearings or replaceable bronze bushings and shall be provided with lubrication means. Unless the ring is of the sealed roller or ball bearing type, the balls or rollers shall be accessible and shall be removable.

3.8.3.2 Hook rollers. Where the holddown function is performed by hook rollers, the rollers may be of the dual purpose type in which one set of rollers performs both the supporting and holddown functions, or may be of the holddown type only. Hook rollers may operate singly or in pairs on tandem bogies and shall be provided with grease fittings. When required for proper operation, hook rollers shall be provided with vertical or horizontal adjustment.

3.8.4 Swing brake. A swing brake, capable of holding the superstructure in any desired position, shall be provided. The swing brake control shall be located within easy reach of the operator, while seated.

3.8.5 Swing clutches. The rotation of the rotating frame shall be controlled through clutches designed for the fast cycling with a clutch capable of transmitting maximum engine torque. The temperature of the swing clutches shall not exceed the maximum temperature recommended by the clutch lining manufacturer.

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3.8.6 Drums. Primary and secondary load hoist drums shall be provided. Pitch diameters of the drum or drum laggins shall be not less than 18 times the diameter of the largest rope used thereon. The drums shall have sufficient rope capacity to accommodate the rope lengths specified in 3.11. Power at the load hoist drum shall be not less than that required to meet lifting capacities specified at recommended reeving. Load brakes shall be capable of holding or smoothly lowering rated loads on recommended reeving, when properly adjusted. The brakes shall be capable of being locked in a set position. Drum clutches shall be capable of withstanding fast cycling continuous operation, as encountered in normal operation in commercial applications, without overheating or excessive wear. Primary and secondary load hoist drums shall have the same pitch diameter. Both primary and secondary load hoist drums shall be provided with drum line turn indicators with the drum rotation information provided to the operator in the operators cab.

3.8.7 Powered load lowering. The primary and the secondary load hoist drums shall be equipped with a powered reversing drive, and suitable control features to provide precision controlled power lowering of loads. The powered load lowering mechanism shall be controlled by a lever conveniently located for the operator within the revolving superstructure cab and shall be capable of lowering all rated loads with the manufacturer's recommended reeving. The powered load lowering mechanism shall be capable of lowering the loads at a smooth, controlled rate of speed, dependent on engine speed, without jerking, bouncing or slipping the load. A braking device may be used to assist the power lowering mechanism. Extended use of the powered load lowering mechanism shall not result in clutch or brake failure by fading, grabbing, cracking, internal rupture, breakage, permanent deformation or any malfunction. Clutch and brake temperatures shall not exceed the maximum recommended by the manufacturer. The powered load lowering mechanism shall be capable of being disengaged to allow load free falling and also be capable of being disengaged, when not in use, so as not to interfere with normal crane operations.

3.8.8 Boom hoist and power lowering. The crane boom hoist shall be power controlled and shall be capable of raising the boom with capacity load. The boom hoist, equipped with a safety ratchet or other positive holding device, shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for power lowering under the operator's control, and shall make gravity lowering (free spooling) of the boom cable impossible under any circumstance. The boom hoist clutch shall be readily accessible for adjustment and repair, and sufficiently large to eliminate slippage under 117.6 percent of rated load.

3.8.9 Operating controls. Controls shall conform to ANSI B30.5, and shall be provided with means to minimize the forces required to actuate the levers in order to minimize operator fatigue in container handling. All brakes shall be capable of being easily locked or unlocked. The operating

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controls shall be identified by a plate indicating the name of the control with arrows or appropriate markings showing the direction of movement for operation.

3.8.10 Lighting and associated electrical system. The lighting system for the crane superstructure shall be either 12 or 24-volt. Electrical components (wire, switches, controls, connectors, light and accessories) shall be suitable for use in a marine atmosphere.

3.8.10.1 Floodlights. Two adjustable floodlights shall be installed near the front and top of the cab, one on each side of the hoist line, and located in a position that the lenses cannot be seen by the operator during normal operations. The floodlights shall be secured in brackets, capable of withstanding severe shock and vibration and permitting horizontal and vertical movement of the light beam and positive clamping of the light in any position. The floodlights shall not extend beyond the top or sides of the cab unless provision is made for an adjustable bracket to fold clear of the cab sides for transport. Floodlamp sealed beam units shall be of the floodlamp type, 60-watt size. Two detachable floodlights shall be installed near the tip end of the boom butt section, one on each side to illuminate area below hook, and located in a position that the lenses cannot be seen by the operator during normal operations. The floodlights shall be secured in brackets, capable of withstanding severe shock and vibration and permitting horizontal and vertical movement of the light beam and positive clamping of the light in any position. Floodlight fixture and bulb units shall be of the floodlamp type of a 500-watt minimum size. Power plant for floodlights shall be integral with the crane. Electrical cable shall be flexible type SO in accordance with UL 62 and with the conductor size as required for the application. Conductors shall be protected from charring and abrasion where they pass through metal enclosures.

3.8.10.2 Interior lights. The superstructure cab and the elevated operators cab shall have not less than one ceiling light with rough service commercial bayonet type lamps with bulbs. The lights shall be located for maximum illumination of the cab interior and so as to provide adequate illumination for access to and exit from the cab. All ceiling lights shall be adequately shaded to eliminate reflected glare from cab windows.

3.8.10.3 Warning horn. A suitable warning horn, electric or air operated, shall be mounted in a protected location on the exterior of the revolving superstructure. The horn shall be operated by a control mounted within reach of the operator.

3.9 Front-end operating equipment. Front-end operating equipment furnished shall be complete with all parts. The following equipment shall be furnished with each crane:

(a) Crane boom.
(b) Crane hook block(s).

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3.9.1 Crane boom. The crane boom shall include the basic boom, center sections, boom safety stop, boom angle indicator and boom hoist clutch throwout.

3.9.1.1 Basic boom and center sections. The basic boom shall not be less than 50 feet, shall consist of the butt (lower) section and a heavy duty tip (upper) section, and shall be complete with all necessary supporting hardware. The basic boom shall be suitable for insertion of center sections to provide for a total boom length of not less than 130 feet. The contractor shall furnish the minimum number of center sections to increase the basic boom in 10 foot increments up to the total length of 130 feet. Center sections shall be provided complete with pendant cables and connecting hardware.

3.9.1.2 Boom safety stop(s). A boom stop(s) shall be furnished in order to provide a safeguard against overtopping of the boom and shall be installed so as to limit the boom angle to not more then 85° above the horizontal plane. The boom stop(s) shall provide a shock absorbing action and shall allow the boom point to be lowered to the ground.

3.9.1.3 Boom hoist clutch throwout. Means shall be provided to disengage the clutch of boom hoist drums when the hoisting boom has reached the maximum angle or height and is in contact with the boom stop(s).

3.9.1.4 Boom angle indicator. A boom angle indicator of the heavy pendulum type swinging on bearings shall be provided. The pointer shall indicate the angle of the boom with the horizontal plane and shall be clearly visible to the operator. A chart shall be provided in the cab, clearly visible to the operator, showing rated lifting capacities at various boom angles for various boom lengths.

3.9.2 Hook blocks. The Type I crane shall be furnished with one hook block rated at not less than 140 tons, and the Type II crane shall be furnished with one hook block rated at not less than 80 tons. Each crane shall be furnished with two hook blocks each rated at not less than 42 tons. The hook blocks shall be of the lubricated type with swivel hook and safety latch. Each block shall be conspicuously marked with its rated capacity.

3.9.3 Load indicating and warning device. A load indicating and warning device of the load-moment type shall be provided to perform the following functions while on outriggers:

(a) Provide a dial or digital reading of hook load in pounds.
(b) Provide a dial or digital reading of radius in feet.
(c) Provide a dial or digital reading of load-moment as a percentage of rated capacity, or a reading of rated capacity.
(d) Provide visual warning (amber light) when load-moment reaches

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90% of rated capacity and visual and audible (red light and alarm) warning at 100% rated capacity.

The device shall provide for manual selection of parts of line, boom length and single line jib operation. The device shall conform to SAF Recommended Practices J159, J675 and J376.

3.10 Sheaves. The pitch diameter of all sheaves shall conform to SAE J881. All sheaves carrying hoist lines shall be guarded to prevent the rope from leaving the sheave groove during operation.

3.11 Ropes. Wire rope conforming to SAE J959 shall be furnished. For each of the load hoist drums, the wire rope shall be the same size and the Type I crane shall be furnished with not less than 1000 feet of wire rope and the Type II crane shall be furnished with not less than 1200 feet of wire rope.

3.11.1 Rope sockets. Ends of standing ropes shall be secured by swaged or speltered socket type fittings. All other ropes shall have ends attached by wedge type sockets.

3.12 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Lubrication fittings shall be in accordance with SAE J534, and shall be located in a protected position accessible to a hand gun. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The crane shall be lubricated prior to delivery with lubricants compatible with military lubricants, MIL-L-10295, MIL-L-10324, or MIL-G-10924. The cranes shall be conspicuously tagged to identify the lubricants and their temperature range.

3.13 Lifting and tiedown provisions and markings. The crane and its components shall be furnished with lifting and tiedown provisions to facilitate disassembly and land and water transport (see 3.5). The provisions shall be either integral and permanently affixed attachments or areas of components specified by the contractor to be used for such purposes. The provisions and markings shall be in accordance with MIL-STD-209, Type III. As certification of meeting the requirements, the contractor shall furnish as part of the first article test report, the structural analysis of the provisions (see 4.3.11).

3.14 Identification marking plates. Each crane shall be provided with a plate, made of noncorrosive metal, 0.035 inch minimum thickness on which is stamped, in letters not less than 0.25 inch high, the following information:

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Nomenclature
NSN
Serial Number
Model
Contract Number
Shipping Date
USA Number

Each front-end attachment and crane boom section furnished by the manufacturer shall be provided with a plate specified above with the following information:

For use with: Crane nomenclature_____
NSN_____
Model_____
Manufacturer_____

3.15 Instruction plates or decals. The crane shall be equipped with suitably located instruction plates or decals describing any special or important procedure to be followed in operating and servicing the equipment. Plates shall be of a noncorrosive metal and shall be permanently mounted on the crane.

3.16 Cleaning, pretreatment and painting. All parts of the crane normally painted shall be cleaned, pretreated and painted in accordance with good commercial practice. The primer and finish coats, the dry film coat thicknesses and the forest green camouflage color shall be in accordance with MIL-T-704.

3.17 Stenciling. The working weight of the basic lifting crane, the weights of the components when disassembled for transport, lifting and tie-down provisions, identification and prescribed tire pressures shall be stenciled in suitable locations and in such a manner as to be readily discernible. Stencil lettering shall be block or stencil-type and shall be not less than 1 inch high.

3.18 Tools and accessory storage. Means shall be provided to carry those tools and accessories required to maintain the crane. Toolbox(es) shall be made of sheet steel of not less than 0.0747 inch nominal thickness (US standard gage number 14). The toolbox(es) shall have a hinged lid and means that will keep the lid closed when subjected to vibration. The toolbox(es) shall be securely fastened to the crane in a protected and accessible location. Drain holes shall be provided to prevent accumulation of water.

3.19 Tools. The contractor shall provide any special tools required for service and maintenance of the cranes (see 6.5).

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3.20 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the crane as requested (see 6.7).

3.21 Workmanship. Workmanship shall be of the highest commercial quality.

3.22 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Unless otherwise specified (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Examination. The first article and each production unit shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's stand product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3 First article tests. When first article is required, the crane shall be tested in accordance with 4.3.1 through 4.3.10. Failure of the crane to pass any test, complete the test, malfunction, incomplete first article, leakage of fluids, breakage, damage, deformation, or inability to meet or exceed the requirements specified in Section 3 shall be the basis for rejection.

4.3.1 Test area. The test area shall be free of obstructions which may cause interference to raising the boom to its maximum height in any position of the rotating base, or interference to rotating the superstructure with the boom in the horizontal position; the surface shall be sufficiently firm to allow passage of the crane without leaving tire tracks deeper than 1/4-inch; a concrete or bituminous surface is desirable; and the test area shall be level within 1 percent in any direction.

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4.3.2 Break-in. The crane shall be operated for a period of not less than 4 hours under a partial load of not less than 1/2 of the rated capacity.

4.3.3 Lifting capacity tests. Lifting capacities of the crane shall be demonstrated by subjecting the crane to load stability tests in accordance with SAE J765. The test shall be conducted on 5 boom lengths including the longest and shortest (see 3.9.1.1) in the least stable and most stable directions; at the minimum working radius, at the radius equal to the boom length, and at three intermediate, equidistant radii, with outriggers. Conformance to 3.4.1 shall be determined by multiplying the balance point load by 85 percent. Where the lifting capacity is based on structural strength or available hoisting power rather than resistance to overturning, the contractor shall submit his test procedure for approval. The procedures shall include all applicable provisions of SAE J765, and shall be designed to show that the crane will lift at least 117.6 percent of the rated capacity loads specified in 3.4.1.

4.3.4 Crane structure stress test. To determine conformance to 3.5.1, the crane contractor shall certify that a complete test has been previously conducted on an identical model, and that the test records are available for inspection. If this certification is not furnished and test records are not available, working stress tests for crane structures shall be conducted in accordance with SAE J987.

4.3.5 Service and parking brake tests. The crane shall be tested in accordance with the applicable Federal Motor Vehicle Safety Standards.

4.3.6 Powered load lowering test. The powered load lowering system shall be tested as follows to determine conformance to 3.8.7:

(a) With the crane equipped with a boom of a length as specified in 3.4.1, and with outriggers set, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum rated load divided by the recommended number of parts of hoist line reeving.
(b) Position the boom angle for the minimum working radius.
(c) Hoist the load 20 feet above the ground level, ± 3 incles, or until interference with the crane boom is obvious.
(d) Operate the power load lowering system in accordance with manufacturer's manual instructions and lower the load under power.
(e) When the load reaches the ground, immediately repeat operations (c) and (d). These operations shall be repeated until 20 cycles have been completed.
(f) Check the temperatures of all linings employed in this operation with at least four thermocouples imbedded within each lining,

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near the working face, and uniformly spaced around the perimeter, or with a contact pyrometer placed directly on the lining.

4.3.7 Swing brake test. Test the swing brake at an engine speed of not less than one half the full load governed speed by engaging the brake with the superstructure in several positions and attempt to rotate the superstructure. Rotation or movement of the superstructure shall constitute failure. Check to insure that the superstructure will remain stationary when the boom is positioned in the boom rack.

4.3.8 Gantry test. The gantry shall be tested to insure conformance to 3.8.2, using a boom length of not less than 130 feet.

4.3.9 Maneuverability test. The crane, equipped with basic boom, boom safety stop, boom angle indicator, and hook block shall be given such tests as are necessary to determine conformance to the travel speeds, gradeability requirements and swing speed as specified in 3.4.3, and that operating controls operate easily.

4.3.10 Sound level measurements. Sound level measurements shall be made in accordance with the following:

(a) At the carrier operator's station, the vehicle noise will be measured at two-thirds governed engine speed in the highest gear, or at governed engine speed with the transmission in neutral in accordance with SAE J919.
(b) At the crane operator's station, the vehicle noise will be measured at three-quarters governed engine speed, hoisting 50 percent of the maximum load for any selected parts-of-line, in accordance with SAE J919.
(c) In order to determine the exterior 85 dB(A) contour, the crane engine shall be operated at full governed speed, and the superstructure shall be continuously swung. Measurements shall be made in accordance with SAE J88a in all directions from the crane.

4.3.11 Lifting and tiedown provisions. To determine conformance to 3.13 the contractor shall furnish a complete structural analysis of the lifting and tiedown provisions. An incomplete analysis or a determination of inadequate provisions will be cause for rejection of the first article.

4.4 Production unit test. Each crane offered for acceptance shall be tested as specified in 4.3.2.

4.5 Inspection of packaging. The preservation, packing and marking shall be examined to determine compliance with the Section 5 requirements.

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5. PACKAGING

5.1 Preservation, packing and marking. Each complete disassembled crane and all items issued with the crane shall be suitably preserved and packed for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with the carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling systems shall be filled with a fresh clean solution of 50% water and 50% antifreeze, conforming to MIL-A-46153. A tag shall be attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcases shall contain the required amount of lubricating oil. Marking on each unpacked item, assembly and shipping container shall reflect as a minimum the item name, national stock number, quantity, contract or purchase order number, appropriate address markings, and any additional markings as may be required by the contract (see 6.2).

6. NOTES

6.1 Intended use. The truck-mounted cranes are primarily intended for use in material handling operation , specifically container handling.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of crane required (see 1.2.1).
(c) When first article is not required (see 3.2).
(d) When first article is to remain in contractor's plant (see 3.2).
(e) When the low ground bearing pressure outrigger floats are not required for Type I crane (see 3.6.1.3).
(f) When the ground pressure for the Type I crane shall not exceed 12 psi (see 3.6.1.3).
(g) When the ground pressure for the Type II crane may exceed 12 psi (see 3.6.1.3).
(h) Technical publications required (see 3.21 and 6.4).
(i) When first article inspection will be performed at another facility (see 4.1).
(j) When additional markings are required for packaging (see 5.1).

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6.3 First-article crane. Any changes or deviations of production cranes from the approved first-article model during production will be subject to the approval of the contracting officer. Approval of the first-article model will not relieve the contractor of his obligation to furnish cranes conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and contractor's maintenance and operation manual to be furnished with each crane.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of cranes.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.7 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification numbers for the crane. The contractor should then be requested by the contracting officer to apply the weight classification number to the crane, and include instructions for changing the number based on the crane configuration in the operation manual.

6.8 Contractor experience. The contracting officer should include requirements for data required to verify the prospective contractor's required experience in producing the crane such as the sales history for the crane, the customer list for the cranes sold commercially or to other Government agencies, and the dealers list that could have been involved in servicing the cranes.

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MIL-C-52967(ME) Cranes, Truck-Mounted, Container Handling

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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